The historical CH density instrument

Controls: Historical CH density index; Historical CH coverage (State level).

1. Describe historical density within a modern state. Let P_{jt} be the population of historical reporting area j in census year t, and a_{jt} its mapped area in square kilometres. A reporting area is usually one historical county; counties with a joint population report form one combined area. Only source areas in the selected historical states or states-and-territories scope enter.

Population and area are split across modern states in proportion to geographic overlap, using the same allocation as area-weighted population. Let s_{jg} be the fraction of source area j inside modern state g. A source entirely inside one state has share one. Splitting leaves the historical source density P_{jt}/a_{jt} unchanged because population and area receive the same share.

2. Average log densities using population shares. Using source areas with known positive population and usable area, the instrument is

$$z_{g,t} = \frac{\sum_j s_{jg} P_{jt} \log(P_{jt}/a_{jt})}{\sum_j s_{jg} P_{jt}}.$$

A source with twice the assigned population receives twice the weight. The index describes log density around an average historical resident, so population concentration matters as well as average state density. Recorded zero-population areas receive zero weight. Unknown populations are omitted from both sums; they are not set to zero.

3. Keep the instrument fixed while estimating the elasticity. At $\theta = 1$, a historical CH factor equals one in every state. Its slope at that point is $z_{g,t}$ and can differ across states: it describes the model's initial productivity response to a small density effect, given historical settlement. We use this slope as a fixed instrument while the modern model estimates $\rho = \theta - 1$. A factor evaluated at a fixed, non-unit θ is another possible instrument, but requires choosing that value beforehand. Our construction avoids this choice; it does not establish instrument validity or strength. The index enters directly, without another logarithm.

4. Choose the coverage requirement. The options are *80%*, *95%*, and *No coverage filter*; the default is 80%. Coverage is the share of the modern state's mapped area assigned to historical sources with known population and usable geometry. Recorded zeros count; unknown populations do not. Split areas count in proportion to the assigned share. The same choice applies in both selected historical years.

No coverage filter removes the area cutoff, but a finite index and positive assigned population remain necessary. A state can then enter through a very small border overlap; inclusion does not imply full historical coverage. Mapped area is polygon area, not a separate land-only measure. Missing records remain flagged. The separate area-weighted population instrument keeps its own 95% rule. Coverage alone establishes neither instrument relevance nor the exclusion restriction.